



# **The Islamia University of Bahawalpur**

## **Project Title: AI-Powered Virtual Try-On System with Real-Time AR and Cloud Integration**

**A project presented to  
Department of Information Technology , IUB Bahawalpur**

### **Project Supervisor**

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**Saeed**

**Roll No: F22BINFT1M01149**

**Session: 2022-26**

**Section: 3(M)**

# DECLARATION

I hereby declare that the Final Year Project entitled **“AI-Powered Virtual Try-On System with Real-Time AR and Cloud Integration”** is the result of my own work and effort carried out under the guidance and supervision of the assigned supervisor. This project, including its software implementation, documentation, analysis, design, and development, has not been copied wholly or partially from any source without proper acknowledgment. All information, references, research materials, and technical resources used during the completion of this project have been appropriately cited and acknowledged where necessary.

It is further declared that this project report and software system have been developed for academic purposes in partial fulfillment of the requirements for the degree of **Bachelor of Information Technology (2022–2026)** at **The Islamia University of Bahawalpur**. No part of this work has previously been submitted to any university, institution, or organization for the award of any degree, diploma, or qualification. I accept full responsibility for the authenticity and originality of this work.

I further declare that neither the whole project nor any part of it has been copied from any unauthorized source or reproduced from another project without proper acknowledgment. Any references, research materials, technologies, or external resources used during development have been properly cited where necessary. No part of this project has been submitted previously to any university or institution for the award of any degree, diploma, or qualification. I accept full responsibility for the originality and authenticity of this work.

## CERTIFICATE OF APPROVAL

Hereby we certify that the above mentioned Final Year Project " **AI-Powered Virtual Try-On System with Real-Time AR and Cloud Integration**" was done by **Muhammad Shifan Saeed** having Roll No. **F22BINFTIM01149** under our supervision and guidance. We have gone through the above mentioned project and found it satisfactory enough for submission for the partial requirement of Bachelors of Science in Information Technology.

### Supervisor

Name : Ms. Kainat

Designation : Lecturer (Department of IT)

Signature: 

Date: 03/06/2026

### External Examiner

Name: \_\_\_\_\_

Designation & Organization: \_\_\_\_\_

Signature: \_\_\_\_\_

Date: \_\_\_\_\_

### Chairman, Department of Information Technology

Name: \_\_\_\_\_

Signature: \_\_\_\_\_

Date: \_\_\_\_\_

Official Stamp: \_\_\_\_\_

# Acknowledgement

First and foremost, I would like to express my sincere gratitude to Allah Almighty for granting me the strength, knowledge, and perseverance required to successfully complete this Final Year Project entitled **“AI-Powered Virtual Try-On System with Real-Time AR and Cloud Integration.”**

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Special thanks go to all my teachers and faculty members whose knowledge and guidance throughout my academic journey helped me build the skills required for this project. Finally, I am especially grateful to my parents, family, and friends for their unwavering support, motivation, patience, and encouragement.

Their belief in my abilities inspired me to overcome challenges and successfully accomplish this project.

**Muhammad Shifan Saeed**  
**F22BINFT1M01149 (3M)**

## ABBREVIATIONS

| Abbreviation | Full Form                           |
|--------------|-------------------------------------|
| AI           | Artificial Intelligence             |
| AR           | Augmented Reality                   |
| API          | Application Programming Interface   |
| CNN          | Convolutional Neural Network        |
| CV           | Computer Vision                     |
| UI           | User Interface                      |
| UX           | User Experience                     |
| SRS          | Software Requirements Specification |
| SDD          | Software Design Document            |
| UML          | Unified Modeling Language           |
| ERD          | Entity Relationship Diagram         |
| MVC          | Model-View-Controller               |
| JWT          | JSON Web Token                      |
| HTTP         | Hypertext Transfer Protocol         |
| HTTPS        | Hypertext Transfer Protocol Secure  |
| REST         | Representational State Transfer     |
| JSON         | JavaScript Object Notation          |
| SQL          | Structured Query Language           |
| DBMS         | Database Management System          |
| GPU          | Graphics Processing Unit            |
| CPU          | Central Processing Unit             |
| AWS          | Amazon Web Services                 |
| GCP          | Google Cloud Platform               |
| CDN          | Content Delivery Network            |
| OpenCV       | Open Source Computer Vision Library |
| ML           | Machine Learning                    |
| DL           | Deep Learning                       |
| FPS          | Frames Per Second                   |
| RGB          | Red, Green, Blue                    |
| URL          | Uniform Resource Locator            |
| IDE          | Integrated Development Environment  |

## CHAPTER 1

# INTRODUCTION

### 1.1 Brief:

The rapid advancement of Artificial Intelligence (AI), Augmented Reality (AR), and Cloud Computing has transformed the way users interact with digital platforms. Online shopping has become increasingly popular; however, customers often face difficulties in visualizing how a product will look before making a purchase. This limitation leads to uncertainty, lower customer satisfaction, and increased product return rates.

The proposed project, **AI-Powered Virtual Try-On System with Real-Time AR and Cloud Integration**, aims to address these challenges by providing an intelligent virtual try-on experience for eyewear and accessories. The system utilizes computer vision and AI-based facial landmark detection to accurately identify facial features and overlay virtual products in real time. AR technology enables realistic visualization, while cloud integration ensures scalability, storage, and efficient system management.

The project has been developed using modern technologies such as React.js, FastAPI, MediaPipe, OpenCV, Three.js, and cloud-based services. The report discusses project analysis, requirements, system design, implementation, testing, and future enhancements.

### 1.2 Relevance to Course Modules:

This project incorporates knowledge acquired from various courses studied during the Bachelor of Information Technology program.

**Artificial Intelligence:** Used for facial landmark detection, object alignment, and intelligent image processing.

**Computer Vision:** Enables face tracking, image analysis, and real-time feature detection.

**Software Engineering:** Applied in requirements gathering, system design, documentation, testing, and project management.

**Database Systems:** Used for storing user information, product data, and session records.

**Web Development:** Supports frontend and backend development using modern web technologies.

**Cloud Computing:** Provides scalable infrastructure, storage, and deployment capabilities.

**Human Computer Interaction (HCI):** Ensures a user-friendly and interactive interface.

### **1.3 Project Background:**

The growth of e-commerce platforms has significantly changed consumer purchasing behavior.

Despite these advancements, customers still face challenges when buying products such as eyewear and accessories online because they cannot physically try products before purchasing.

Traditional online shopping platforms rely on static images and product descriptions, which often fail to provide a realistic understanding of product appearance. As a result, customers experience uncertainty, leading to decreased confidence in purchasing decisions.

Recent developments in AI, AR, and computer vision technologies have enabled the creation of virtual try-on systems that allow users to visualize products directly on their faces using cameras or uploaded images. This project leverages these technologies to create an intelligent virtual try-on platform that enhances user experience and reduces product return rates.

### **1.4 Related Material and Literature:**

Several researchers and organizations have developed virtual try-on technologies to improve online shopping experiences.

Major e-commerce companies such as Amazon, Lenskart, and Warby Parker have integrated virtual try-on features into their platforms. These systems utilize computer vision and augmented reality technologies to provide realistic product visualization.

Research studies indicate that virtual try-on systems significantly improve customer engagement, purchasing confidence, and overall shopping satisfaction. Recent advancements in MediaPipe, OpenCV, and deep learning frameworks have further improved the accuracy and efficiency of facial landmark detection and AR rendering.

Cloud-based architectures have also become increasingly important for handling large-scale user interactions and ensuring system scalability.

### **1.5 Analysis from Literature Review (in the context of the project):**

The literature review demonstrates that existing virtual try-on solutions provide valuable functionality but often face limitations related to rendering accuracy, system scalability, and device compatibility.

Many existing systems require high-end hardware or expensive software solutions, making them inaccessible to a broader audience. Some solutions focus primarily on cosmetic products or eyewear while lacking flexibility for additional accessory categories.

The proposed AI-Powered Virtual Try-On System addresses these challenges by integrating lightweight AI models, efficient AR rendering, and cloud-based infrastructure. The use of MediaPipe and modern web technologies ensures compatibility across multiple devices while maintaining real-time performance and accuracy.

The project aims to provide a cost-effective, scalable, and user-friendly solution suitable for both academic research and commercial applications.

## **1.6 Methodology and Software Lifecycle for This Project:**

The project follows an Agile-inspired Software Development Life Cycle (SDLC) combined with a modular development methodology.

The development process consists of the following phases:

1. Requirement Gathering and Analysis
2. System Design
3. Development and Implementation
4. Testing and Validation
5. Deployment
6. Maintenance and Future Enhancements

The modular approach enables independent development of system components such as the AI Engine, AR Renderer, User Management Module, and Cloud Services Module.

### **1.6.1 Rationale behind Selected Methodology:**

The Agile methodology was selected because it supports iterative development, continuous improvement, and flexibility in handling changing requirements.

The virtual try-on system involves multiple technologies, including AI, AR, cloud computing, and web development. Agile development allows features to be implemented and tested incrementally, reducing development risks and improving project quality.

Additionally, the modular architecture enhances maintainability, scalability, and future extensibility. Each component can be independently updated without affecting the overall system, making the solution more reliable and easier to manage.

The selected methodology ensures efficient resource utilization, better project monitoring, and successful delivery of project objectives within the academic development timeline.

## CHAPTER 2

### PROBLEM DEFINITION

This chapter defines the problem addressed by the proposed system and describes the expected deliverables, development requirements, and analysis of the current system. It highlights the limitations of existing online shopping experiences and explains how the proposed AI-Powered Virtual Try-On System provides an effective solution.

#### **2.1 Problem Statement:**

The rapid growth of e-commerce platforms has transformed the retail industry by enabling customers to purchase products online conveniently. However, customers often face difficulties when purchasing eyewear and accessories because they cannot physically try the products before making a purchase. Traditional online stores typically provide only static images and textual descriptions, which do not accurately represent how a product will appear on a user.

As a result, customers frequently experience uncertainty regarding product appearance, size, fitting, and suitability. This uncertainty leads to lower customer confidence, reduced purchasing decisions, increased product return rates, and financial losses for businesses.

Existing virtual try-on solutions are often expensive, hardware-dependent, or limited in functionality. Many systems require specialized devices or fail to provide accurate real-time visualization.

The primary objective of this project is to develop an **AI-Powered Virtual Try-On System with Real-Time AR and Cloud Integration** that enables users to virtually try eyewear and accessories using live camera feeds or uploaded images. The system aims to provide realistic visualization, improve customer confidence, reduce return rates, and enhance the overall online shopping experience through Artificial Intelligence, Computer Vision, Augmented Reality, and Cloud Computing technologies.

#### **2.2 Deliverables and Development Requirements:**

##### **Project Deliverables**

The following deliverables will be produced upon successful completion of the project:

##### **1. Web-Based Virtual Try-On Application**

A responsive web application that allows users to access the virtual try-on system through modern web browsers.

##### **2. User Authentication Module**

A secure registration and login system that manages user accounts and access permissions.

##### **3. Product Catalog Management System**

A module for storing, displaying, searching, and managing eyewear and accessory products.

##### **4. AI-Based Face Detection Module**

An intelligent module that detects facial landmarks and identifies face positions using computer vision techniques.

## **5. AR Visualization Module**

A real-time augmented reality system that overlays virtual products accurately on the user's face.

## **6. Image Upload Feature**

A functionality that allows users to upload images and perform virtual try-on operations.

## **7. Admin Dashboard**

An administrative panel for managing products, users, analytics, and system configurations.

## **8. Database System**

A database for storing user information, product details, virtual try-on sessions, and application records.

## **9. Cloud Integration**

Cloud services for data storage, scalability, system monitoring, and future deployment.

## **10. Project Documentation**

Complete project documentation including:

- Software Requirements Specification (SRS)
- Software Design Document (SDD)
- Final Project Report
- User Manual
- Testing Documentation

## **Development Requirements**

### **Hardware Requirements:**

- Intel Core i5 Processor or higher
- Minimum 8 GB RAM
- Webcam or smartphone camera
- Internet connection
- Storage space for application deployment

### **Software Requirements:**

- Operating System: Windows 10/11 or Linux
- Frontend: React.js
- Backend: FastAPI
- Database: PostgreSQL or MongoDB
- Computer Vision: OpenCV
- AI Framework: MediaPipe
- AR Rendering: Three.js
- Cloud Platform: AWS or equivalent cloud service
- Development Tools: Visual Studio Code, Git, GitHub

### **2.3 Current System:**

Currently, most online shopping platforms rely on static product images and descriptions to present products to customers. While these methods provide basic product information, they do not offer users a realistic understanding of how a product will appear when worn.

For eyewear and accessories, customers often struggle to determine:

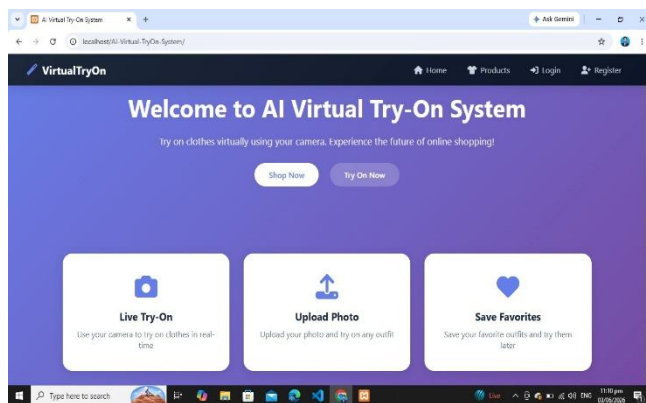
- Whether the product suits their facial features.
- Whether the size and shape are appropriate.
- How the product will look from different viewing angles.
- Whether the product matches their personal style.

As a result, customers frequently purchase unsuitable products and later return them. This increases operational costs for businesses and decreases customer satisfaction.

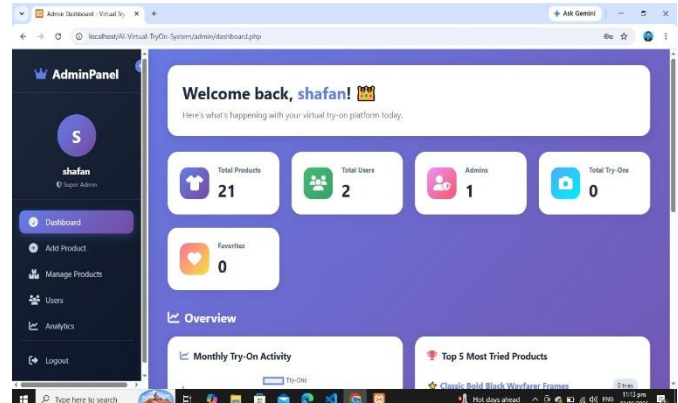
Some commercial platforms have implemented virtual try-on technologies; however, many of these solutions are proprietary, expensive, limited to specific devices, or unavailable to smaller businesses. Additionally, certain systems require specialized hardware and may not provide sufficient accuracy or scalability.

The proposed system overcomes these limitations by combining AI-powered facial landmark detection, real-time augmented reality rendering, and cloud-based infrastructure. Users can instantly visualize eyewear and accessories on their faces through live camera feeds or uploaded images, providing a more interactive and reliable shopping experience.

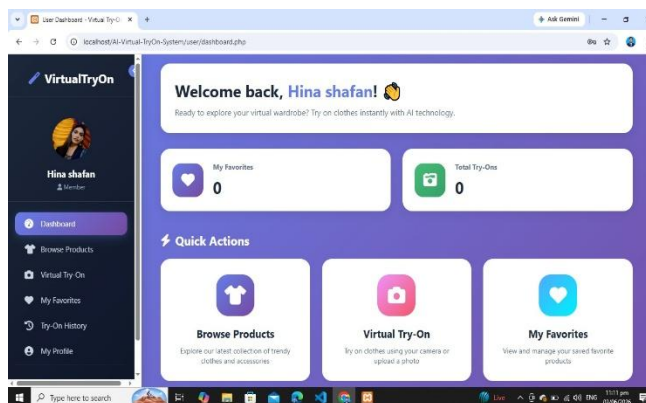
**Figure 2.1: Current Online Shopping Process**



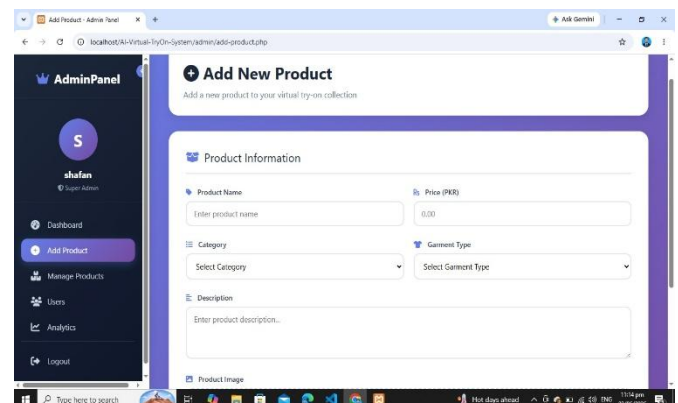
**Homepage Layout**



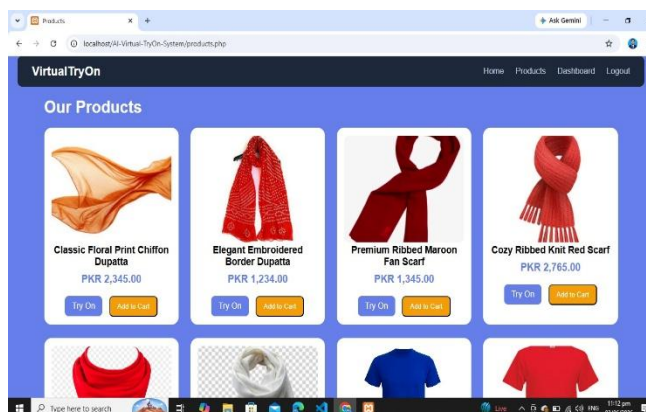
**Admin Pannel**



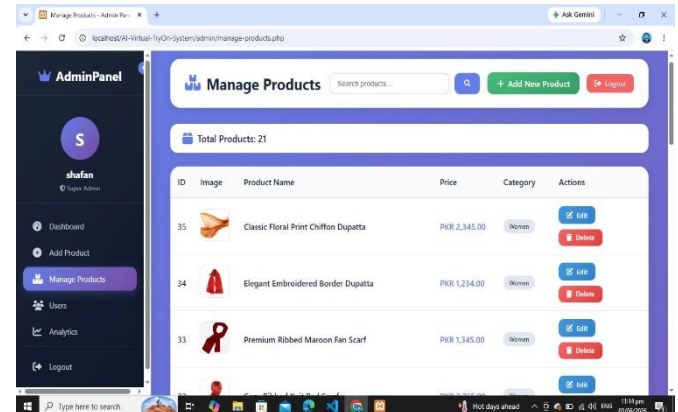
**User Pannel**



**Products Listing**



**Browse Products**



**Products Managed**

A flow diagram illustrating the traditional online shopping process:

Customer → Browse Product Images → Read Product Description → Purchase Decision → Product Delivery → Possible Return

This figure demonstrates the absence of product visualization capabilities in traditional e-commerce systems and highlights the need for an intelligent virtual try-on solution.

# **(SRS): Software Requirement Specifications**

## **Introduction:**

The e-commerce industry faces significant challenges with high product return rates, particularly for products that customers cannot physically try before purchasing. Virtual try-on technology represents a transformative solution that bridges the gap between online and offline shopping experiences.

This project proposes the development of a cloud-based virtual try-on system that leverages computer vision, deep learning, and augmented reality technologies to provide realistic product visualization. The system enables users to see how products, specifically eyewear and accessories, look on them before making a purchase decision.

The virtual try-on system will support two primary modes of operation: real-time camera mode for live visualization and photo upload mode for trying products on saved images. Both modes will provide accurate product placement with proper scaling, lighting adjustment, and perspective transformation.

## **Purpose:**

The primary goals and objectives of this project are:

- To develop a real-time virtual try-on system using computer vision and AR for eyewear and accessories
- To implement accurate face detection with 68 facial landmark identification
- To create realistic product overlay with proper scaling, lighting, shadows, and perspective adjustment
- To provide size recommendations based on facial measurements
- To build a responsive web and mobile interface with live camera and photo upload modes
- To integrate cloud infrastructure for scalability, storage, and multi-user support
- To implement real-time processing with minimal latency (under 200ms per frame)
- To create a product catalog management system with 3D model storage
- To generate analytics on user behavior, popular products, and try-on patterns
- To achieve realistic rendering indistinguishable from actual product photos

## **Scope:**

The following items are within the scope of this project:

- Virtual try-on for eyewear (glasses, sunglasses) and accessories
- Real-time face detection and tracking using MediaPipe
- 3D product rendering using Three.js
- Web application using React.js with camera access
- Cloud deployment with user authentication
- Product catalog management with 3D models
- Size recommendation engine
- User analytics and behavior tracking

The following items are outside the scope of this project:

- Full-body clothing try-on
- Physical inventory management
- Payment processing and e-commerce transactions
- AR try-on in physical stores
- VR headset support
- Voice command interaction

## **Product Perspective:**

The Virtual Try-On System is designed as a standalone cloud-based application that can be integrated with existing e-commerce platforms through REST APIs. The system follows a multi-tier architecture:

- Frontend Layer: Responsive web interface accessible from any device with camera support
- Processing Layer: Computer vision and 3D rendering services for real-time product visualization
- Backend Layer: REST APIs for user management, product catalog, and analytics
- Cloud Infrastructure: Scalable cloud storage, database, and compute resources

The system can function as an independent application or be embedded into existing ecommerce websites through white-label API integration.

## **User Characteristics:**

The system will serve the following user types:

- End Users/Customers: Primary users who want to try on products virtually before purchasing. They may have varying levels of technical expertise and access the system through web browsers or mobile devices.
- Store Administrators: Business users who manage the product catalog, upload 3D models, and monitor analytics. They require access to the admin dashboard.
- System Administrators: Technical staff responsible for system maintenance, monitoring, and configuration.
- E-commerce Integration Developers: Third-party developers who integrate the virtual try-on functionality into their platforms using the provided APIs.

## **Similar Apps and Systems / Literature Review:**

Several virtual try-on solutions exist in the market, each with their own strengths and limitations:

Warby Parker Virtual Try-On: Allows users to try on eyewear using their phone camera. Strengths include smooth user experience and accurate face tracking. Limitations include being restricted to their own product catalog and requiring a mobile app.

Ray-Ban Virtual Try-On: Web-based try-on for sunglasses. Offers good lighting adjustment but has limited product customization options and occasional tracking issues in low light.

Lenskart 3D Try-On: Popular in South Asian markets with extensive eyewear catalog. Provides good product variety but sometimes struggles with accurate sizing recommendations.

Our proposed system aims to improve upon existing solutions by providing: more accurate face tracking with 68 landmark points, better lighting and shadow simulation, ML-based size recommendations with 85%+ accuracy, and a scalable cloud architecture supporting white-label integration.

## **Proposed Technologies:**

The following technologies will be used to develop this system:

### **Programming Languages:**

- Python: Backend services, machine learning models, computer vision processing
- JavaScript/TypeScript: Frontend web application, 3D rendering

### **Computer Vision and AI:**

- OpenCV: Image processing and manipulation
- Media Pipe: Face and hand detection with 68 landmark points
- TensorFlow/PyTorch: Deep learning models for size recommendation
- Dlib: Facial landmark detection backup

### **AR and 3D Rendering:**

- Three.js: 3D rendering in web browser
- AR.js: Augmented reality features
- Blender: 3D model creation and editing

### **Frontend:**

- React.js: Web application interface
- Tailwind CSS: UI styling
- WebRTC: Real-time video streaming

### **Backend:**

- Fast API: Python REST API framework
- WebSocket: Real-time communication
- Redis: Caching for fast performance

### **Cloud Infrastructure:**

- AWS/Google Cloud: Cloud platform
- AWS S3/Cloud Storage: Product models and images
- MongoDB/PostgreSQL: User data and product catalog
- Docker: Containerization
- JWT: Authentication

## **Requirements:**

The Virtual Try-On System provides a comprehensive solution for online product visualization. The core functionality enables users to see how eyewear and accessories would look on them through either real-time camera feed or uploaded photos. The system processes video frames to detect faces, identify facial landmarks, and overlay 3D product models with accurate positioning, scaling, and lighting.

Users can browse a product catalog, select items to try on, and switch between different products in real-time. The system provides size recommendations based on facial measurements extracted from the camera feed. Users can save their favorite products, capture try-on images, and share them on social media. Business users can manage the product catalog through an admin dashboard, upload new 3D models, and view analytics on product popularity and user engagement.

## Functional Requirements:

### FR001 - User Registration

|                |                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>    | FR001                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Purpose</b> | User registration allows new users to create an account to access personalized features including saved try-ons, favorites, and size profiles.                                                                                                                                                                                                                                                                       |
| <b>User(s)</b> | Customer, Admin                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Input</b>   | <ul style="list-style-type: none"><li>• Full Name: User's full name for profile display</li><li>• Email: Valid email address for account verification and communication</li><li>• Password: Minimum 8 characters with at least one number and special character</li><li>• Profile Picture (Optional): User's photo for profile</li><li>• Face Scan (Optional): Initial face scan for size profile creation</li></ul> |
| <b>Output</b>  | <ul style="list-style-type: none"><li>• Account created successfully</li><li>• Verification email sent to user</li><li>• User redirected to onboarding flow</li></ul>                                                                                                                                                                                                                                                |

### FR002 - User Login

|                |                                                                                                                                                                              |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>    | FR002                                                                                                                                                                        |
| <b>Purpose</b> | Registered users can log into their accounts to access saved preferences and try-on history.                                                                                 |
| <b>User(s)</b> | Customer, Admin                                                                                                                                                              |
| <b>Input</b>   | <ul style="list-style-type: none"><li>• Email: Registered email address</li><li>• Password: Account password</li></ul>                                                       |
| <b>Output</b>  | <ul style="list-style-type: none"><li>• User authenticated and session created</li><li>• JWT token generated for API access</li><li>• User redirected to dashboard</li></ul> |

## FR003 Real-Time Camera Try-On

|                |                                                                                                                                                                                                                                        |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>    | FR003                                                                                                                                                                                                                                  |
| <b>Purpose</b> | Users can try on products in real-time using their device camera with live face tracking and product overlay.                                                                                                                          |
| <b>User(s)</b> | Customer                                                                                                                                                                                                                               |
| <b>Input</b>   | <ul style="list-style-type: none"><li>• Camera Access Permission: User grants camera access</li><li>• Selected Product: Product chosen from catalog</li><li>• Lighting Conditions: Ambient lighting detected automatically</li></ul>   |
| <b>Output</b>  | <ul style="list-style-type: none"><li>• Live video feed with product overlay</li><li>• Real-time face tracking at 30fps</li><li>• Proper lighting and shadow simulation</li><li>• Size adjustment based on face measurements</li></ul> |

## FR004 - Photo Upload Try-On

|                |                                                                                                                                                                                      |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>    | FR004                                                                                                                                                                                |
| <b>Purpose</b> | Users can upload a photo to try on products without using live camera.                                                                                                               |
| <b>User(s)</b> | Customer                                                                                                                                                                             |
| <b>Input</b>   | <ul style="list-style-type: none"><li>• Photo File: JPG, PNG, or WebP format, max 10MB</li><li>• Selected Product: Product chosen from catalog</li></ul>                             |
| <b>Output</b>  | <ul style="list-style-type: none"><li>• Photo processed with face detection</li><li>• Product overlay applied to photo</li><li>• Ability to download or share result image</li></ul> |

## FR005 - Product Catalog Browse

|                |                                                                                                                                                                                                                |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>    | FR005                                                                                                                                                                                                          |
| <b>Purpose</b> | Users can browse available products with filtering and search capabilities.                                                                                                                                    |
| <b>User(s)</b> | Customer, Admin                                                                                                                                                                                                |
| <b>Input</b>   | <ul style="list-style-type: none"><li>• Search Query (Optional): Text search for products</li><li>• Filters: Category, brand, price range, color</li><li>• Sort Option: By popularity, price, newest</li></ul> |
| <b>Output</b>  | <ul style="list-style-type: none"><li>• List of matching products displayed</li><li>• Product thumbnails and basic info shown</li><li>• Pagination for large result sets</li></ul>                             |

## FR006 Size Recommendation

|             |       |
|-------------|-------|
| <b>Name</b> | FR006 |
|-------------|-------|

|                |                                                                                                                                                                                     |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Purpose</b> | System provides size recommendations based on facial measurements extracted from camera or photo.                                                                                   |
| <b>User(s)</b> | Customer                                                                                                                                                                            |
| <b>Input</b>   | <ul style="list-style-type: none"> <li>• Face Image: From camera or uploaded photo</li> <li>• Product Category: Type of product being tried</li> </ul>                              |
| <b>Output</b>  | <ul style="list-style-type: none"> <li>• Recommended size (S, M, L or numeric)</li> <li>• Confidence score for recommendation</li> <li>• Alternative sizes if applicable</li> </ul> |

## FR007 - Save to Favorites

|                |                                                                                                                                                                    |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>    | FR007                                                                                                                                                              |
| <b>Purpose</b> | Logged-in users can save products to their favorites list for later reference.                                                                                     |
| <b>User(s)</b> | Customer                                                                                                                                                           |
| <b>Input</b>   | <ul style="list-style-type: none"> <li>• Product ID: Identifier of product to save</li> <li>• User Authentication: Valid login session</li> </ul>                  |
| <b>Output</b>  | <ul style="list-style-type: none"> <li>• Product added to user's favorites</li> <li>• Confirmation message displayed</li> <li>• Favorites count updated</li> </ul> |

## FR008 - Capture and Share Try On

|                |                                                                                                                                                                                        |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>    | FR008                                                                                                                                                                                  |
| <b>Purpose</b> | Users can capture their try-on image and share it on social media or download it.                                                                                                      |
| <b>User(s)</b> | Customer                                                                                                                                                                               |
| <b>Input</b>   | <ul style="list-style-type: none"> <li>• Try-On Session: Active try-on with product overlay</li> <li>• Share Platform (Optional): Facebook, Instagram, Twitter, WhatsApp</li> </ul>    |
| <b>Output</b>  | <ul style="list-style-type: none"> <li>• High-resolution image captured</li> <li>• Image saved to device or shared to selected platform</li> <li>• Shareable link generated</li> </ul> |

## FR009 Admin Product Management

|                |                                                                                                                                                                                                                                                               |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>    | FR009                                                                                                                                                                                                                                                         |
| <b>Purpose</b> | Administrators can add, edit, and remove products from the catalog.                                                                                                                                                                                           |
| <b>User(s)</b> | Admin                                                                                                                                                                                                                                                         |
| <b>Input</b>   | <ul style="list-style-type: none"> <li>• Product Details: Name, description, category, price</li> <li>• 3D Model File: GLB or GLTF format</li> <li>• Product Images: Multiple angles</li> <li>• Size Information: Available sizes and measurements</li> </ul> |
| <b>Output</b>  | <ul style="list-style-type: none"> <li>• Product added/updated in catalog</li> <li>• 3D model processed and stored</li> <li>• Product available for try-on</li> </ul>                                                                                         |

## FR010 - Analytics Dashboard

|                |                                                                                                                                                                                                            |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>    | FR010                                                                                                                                                                                                      |
| <b>Purpose</b> | Administrators can view analytics on user behavior, popular products, and try-on patterns.                                                                                                                 |
| <b>User(s)</b> | Admin                                                                                                                                                                                                      |
| <b>Input</b>   | <ul style="list-style-type: none"> <li>• Date Range: Period for analytics</li> <li>• Metrics Selection: Views, try-ons, favorites, shares</li> </ul>                                                       |
| <b>Output</b>  | <ul style="list-style-type: none"> <li>• Charts and graphs displaying metrics</li> <li>• Top products by try-on count</li> <li>• User engagement statistics</li> <li>• Conversion tracking data</li> </ul> |

# **Non-Functional Requirements:**

## **Performance Requirements:**

- Real-time processing speed of under 200ms per frame
- Smooth video rendering at 30fps minimum
- Page load time under 3 seconds
- API response time under 500ms
- Support for 100+ concurrent users

## **Security Requirements:**

- All data transmitted over HTTPS/TLS
- Passwords hashed using bcrypt with salt
- JWT tokens for API authentication with 24-hour expiry
- Input validation and sanitization on all user inputs
- GDPR-compliant data handling and privacy

## **Usability Requirements:**

- Intuitive user interface requiring no training
- Mobile-responsive design for all screen sizes
- Works on Chrome, Firefox, Safari, and Edge browsers
- Graceful degradation for older browsers
- Clear error messages and user feedback

## **Reliability Requirements:**

- 99% system uptime
- Face detection accuracy above 95%
- Size recommendation accuracy above 85%
- Automatic failover for cloud services
- Daily automated backups

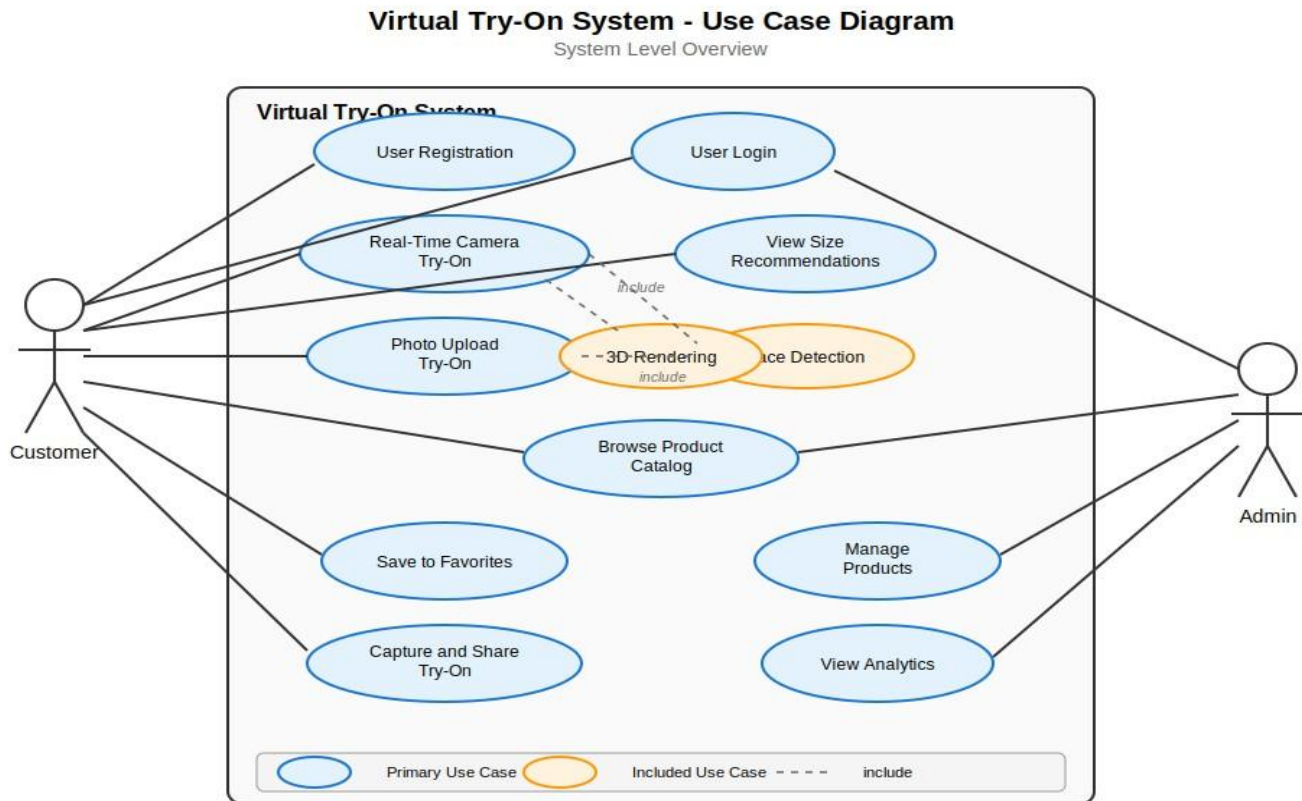
## **Scalability Requirements:**

- Horizontal scaling for increased load
- Auto-scaling based on traffic patterns
- Support for 50+ products in catalog
- CDN for fast global content delivery

## Use Cases and Flow of Processes:

Use cases provide formal representation of process flows defined by functional requirements. The following section presents the system-level use case diagram and detailed use case specifications for key system functions.

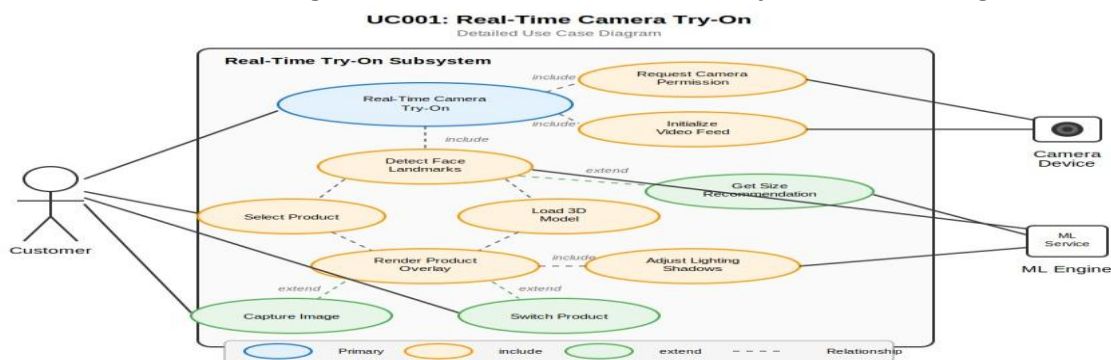
**Figure 1: System Level Use Case Diagram**



## Use Case 1: Real-Time Camera Try-On

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>               | UC001                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Name</b>             | Real-Time Camera Try-On                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Description</b>      | This case describes the process flow for a user trying on products using their device camera in real-time.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Requirement(s)</b>   | FR003, FR006                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Actor(s)</b>         | Customer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Precondition</b>     | User has granted camera permission and has a supported browser                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Postcondition</b>    | Users see live video feed with selected product overlaid on their face                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Basic Flow</b>       | <ol style="list-style-type: none"> <li>1. User navigates to Try-On page</li> <li>2. System requests camera permission</li> <li>3. User grants camera access</li> <li>4. System initializes camera feed</li> <li>5. System detects user's face and identifies 68 landmark points</li> <li>6. User selects a product from the catalog</li> <li>7. System loads 3D model of selected product</li> <li>8. System calculates position, scale, and rotation based on facial landmarks</li> <li>9. System renders product overlay on video feed</li> <li>10. System adjusts lighting and shadows to match environment</li> <li>11. User can switch products or adjust positioning</li> <li>12. User can capture image of try-on result</li> </ol> |
| <b>Alternative Flow</b> | <ol style="list-style-type: none"> <li>1. User declines camera permission - System prompts to use Photo Upload mode instead</li> <li>2. Face not detected - System displays message asking user to adjust position or lighting</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Exceptions</b>       | <ol style="list-style-type: none"> <li>1. Camera not available or not supported</li> <li>2. Poor lighting conditions affecting face detection</li> <li>3. Network interruption during 3D model loading</li> <li>4. Browser not supporting WebRTC</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

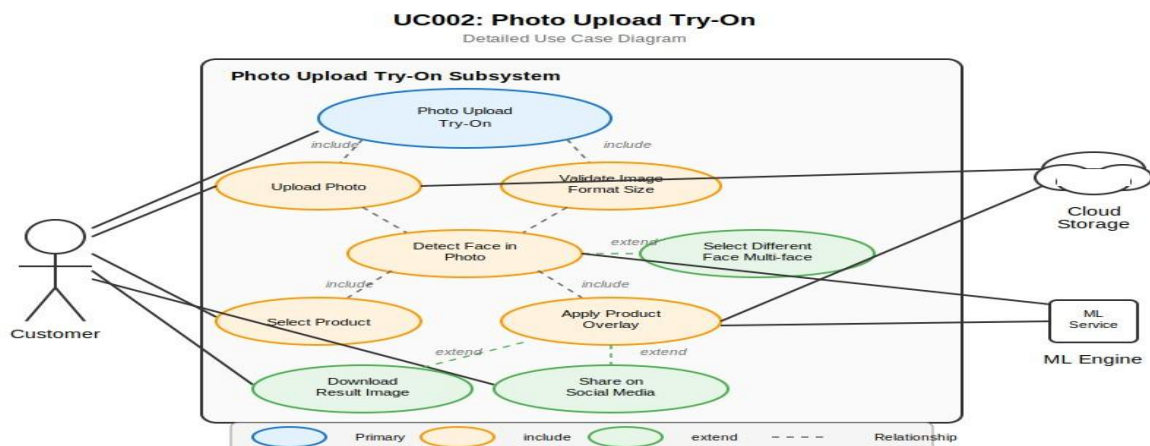
**Figure 2: UC001 - Real-Time Camera Try-On Use Case Diagram**



## Use Case 2: Photo Upload Try-On

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>               | UC002                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Name</b>             | Photo Upload Try-On                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Description</b>      | This use case describes the process flow for a user trying on products by uploading a photo.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Requirement(s)</b>   | FR004, FR006                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Actor(s)</b>         | Customer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Precondition</b>     | User has a photo with a clearly visible face                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Postcondition</b>    | User sees uploaded photo with selected product overlaid on their face                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Basic Flow</b>       | <ol style="list-style-type: none"> <li>1. User navigates to Photo Upload section</li> <li>2. User clicks upload button or drags photo to upload area</li> <li>3. User selects photo from device (JPG, PNG, WebP, max 10MB)</li> <li>4. System uploads and processes the photo</li> <li>5. System detects face in photo and identifies facial landmarks</li> <li>6. System displays photo with face detection confirmation</li> <li>7. User selects a product from the catalog</li> <li>8. System loads 3D model and calculates positioning</li> <li>9. System renders product overlay on photo</li> <li>10. System adjusts lighting to match photo</li> <li>11. User can download or share the result image</li> </ol> |
| <b>Alternative Flow</b> | <ol style="list-style-type: none"> <li>1. Multiple faces detected - System prompts user to select which face to use</li> <li>2. Low quality image - System suggests uploading a higher resolution photo</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Exceptions</b>       | <ol style="list-style-type: none"> <li>1. No face detected in uploaded photo</li> <li>2. File size exceeds 10MB limit</li> <li>3. Unsupported file format</li> <li>4. Photo orientation incorrect</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

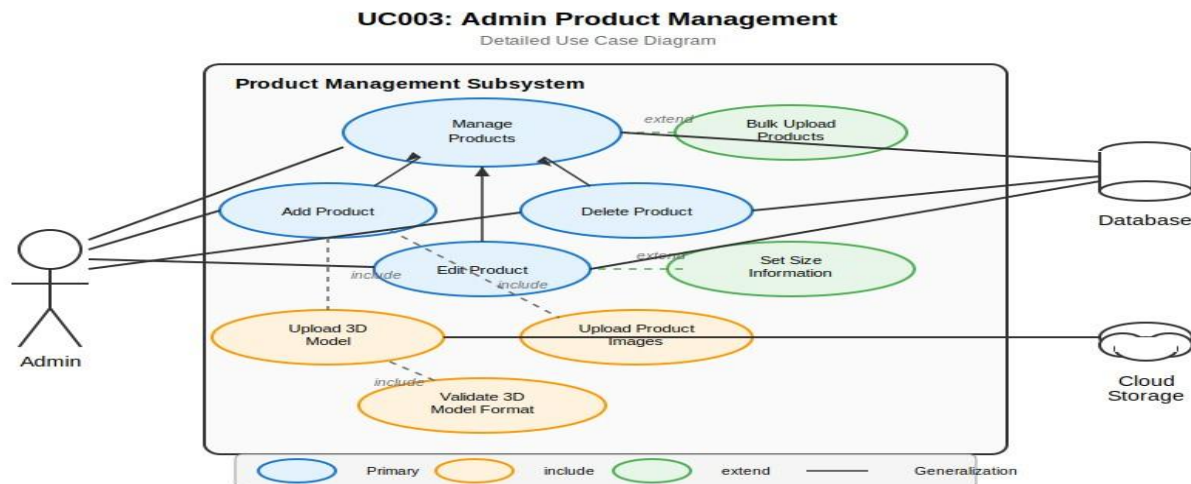
Figure 3: UC002 - Photo Upload Try-On Use Case Diagram



## Use Case 3: Admin Product Management

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>               | UC003                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Name</b>             | Admin Product Management                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Description</b>      | This use case describes the process flow for administrators adding new products to the catalog.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Requirement(s)</b>   | FR009                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Actor(s)</b>         | Admin                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Precondition</b>     | Admin is logged in with administrator privileges                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Postcondition</b>    | New product is added to catalog and available for virtual try-on                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Basic Flow</b>       | <ol style="list-style-type: none"> <li>1. Admin navigates to Admin Dashboard</li> <li>2. Admin clicks 'Add New Product' button</li> <li>3. Admin fills in product details (name, description, category, price)</li> <li>4. Admin uploads product images (multiple angles)</li> <li>5. Admin uploads 3D model file (GLB/GLTF format)</li> <li>6. System validates 3D model format and structure</li> <li>7. Admin specifies size options and measurements</li> <li>8. Admin sets product status (active/inactive)</li> <li>9. Admin clicks 'Save Product'</li> <li>10. System processes and stores 3D model in cloud storage</li> <li>11. System creates product entry in database</li> <li>12. System confirms product added successfully</li> </ol> |
| <b>Alternative Flow</b> | <ol style="list-style-type: none"> <li>1. Bulk upload - Admin uploads CSV with multiple products</li> <li>2. Edit existing product - Admin modifies existing product details</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Exceptions</b>       | <ol style="list-style-type: none"> <li>1. Invalid 3D model format</li> <li>2. Missing required product fields</li> <li>3. 3D model file too large</li> <li>4. Database connection error</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

Figure 4: UC003 - Admin Product Management Use Case Diagram



# Software Design Document

## Introduction

### Purpose of the Document

This Software Design Document (SDD) presents a comprehensive and detailed design specification for the **AI-Powered Virtual Try-On System with Real-Time AR and Cloud Integration**. The primary purpose of this document is to describe the system architecture, components, data structures, interfaces, and design decisions in a clear and professional manner. It is intended to support academic evaluation, system development, future maintenance, and scalability planning. This document serves as a technical reference for developers, instructors, examiners, and other stakeholders involved in the project lifecycle.

### Scope of the System

The scope of this system covers the design and implementation of an intelligent virtual try-on platform that enables users to visualize products such as clothing, accessories, and eyewear in real time using their device camera. The system integrates Artificial Intelligence (AI) for image processing and body/face detection, Augmented Reality (AR) for realistic overlay and visualization, and Cloud Computing for data storage, processing, and scalability. The system is designed to operate as a web-based or mobile-based application and can be extended for e-commerce platforms in the future.

### Definitions, Acronyms, and Abbreviations

| Term | Description                       |
|------|-----------------------------------|
| AI   | Artificial Intelligence           |
| AR   | Augmented Reality                 |
| SDD  | Software Design Document          |
| MVC  | Model View Controller             |
| API  | Application Programming Interface |
| UI   | User Interface                    |
| UX   | User Experience                   |
| CNN  | Convolutional Neural Network      |

## System Overview

The AI-Powered Virtual Try-On System is an intelligent, cloud-enabled application that provides users with an immersive virtual try-on experience. The system captures live images or video streams from the user's device, processes them using AI-based computer vision algorithms, and renders virtual products using AR technology. Cloud services are utilized to manage user data, product models, analytics, and system scalability.

## User Classes and Characteristics

| User Class      | Description                                                                                   |
|-----------------|-----------------------------------------------------------------------------------------------|
| End User        | Individuals who use the system to virtually try products and view results in real time        |
| Administrator   | Authorized personnel responsible for managing products, users, system settings, and analytics |
| System Services | Background services responsible for AI inference, AR rendering, and cloud operations          |

## Design Considerations

### Assumptions

- Users possess a camera-enabled device such as a smartphone, tablet, or webcam-enabled computer
- The system has access to pre-trained AI models for detection and alignment
- Internet connectivity is available for cloud communication

### Constraints

- Real-time performance constraints for AR rendering
- Device hardware limitations such as camera quality and processing power • Network latency and bandwidth limitations

## Design Goals

- High accuracy in product alignment and visualization
- Low latency for real-time interaction
- Scalable and maintainable system architecture
- Secure handling of user data

## Design Methodology

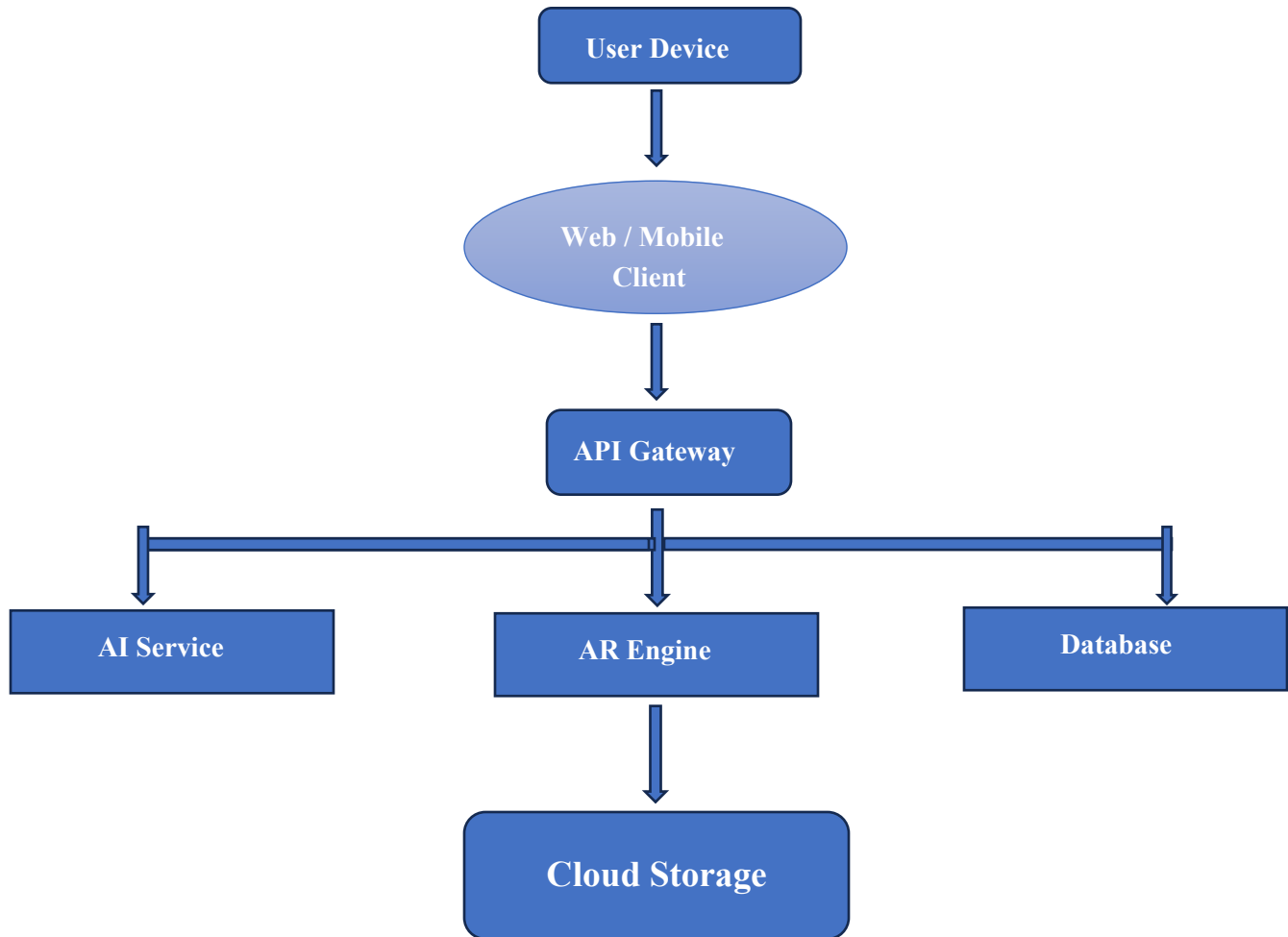
The system follows a **modular, MVC-based architecture** combined with **microservices**. Each major functionality such as AI processing, AR rendering, and data management is encapsulated in independent modules to enhance maintainability and scalability.

## System Architecture

### Architectural Overview

The system adopts a **Client–Server Architecture** integrated with **Cloud-Based Services**. The client handles user interaction and real-time visualization, while the server manages business logic, AI processing, and persistent storage.

#### High-Level Architecture Diagram



### Architectural Style

- MVC pattern for separation of concerns
- Microservices architecture for AI and AR modules
- Cloud-native deployment for scalability and availability

# Data Design

## Database Overview

The system uses a relational database to store structured data such as user profiles and product metadata, while cloud object storage is used for storing images, videos, and 3D models.

### Data Entities and Descriptions

| Entity       | Description                                    |
|--------------|------------------------------------------------|
| User         | Stores personal and authentication information |
| Product      | Stores product details and 3D model references |
| TryOnSession | Stores session data for each virtual try-on    |
| Admin        | Stores administrative credentials and roles    |

## ER Diagram (Textual Representation)

User (user\_id) ---< TryOnSession >--- Product(product\_id)

## Data Integrity and Validation

- Primary and foreign key constraints are enforced
- Input validation is applied at both client and server levels

## Component Design

### Major System Components

| Component    | Responsibility                                               |
|--------------|--------------------------------------------------------------|
| User Module  | Handles authentication, registration, and profile management |
| AI Engine    | Performs image processing, detection, and fitting            |
| AR Renderer  | Renders virtual products in real time                        |
| Admin Module | Manages system configuration and content                     |
| Cloud Module | Handles storage, analytics, and scalability                  |

# AI Engine Design

The AI Engine is responsible for analyzing images using deep learning models such as CNNs. It performs facial landmark detection, body pose estimation, and product alignment to ensure accurate try-on results.

# AR Rendering Module

The AR module overlays 3D product models onto the user’s live camera feed. It continuously tracks movement and adjusts rendering in real time to provide a realistic experience.

# Interface Design

## User Interface Design Principles

- Clean and minimalistic layout
- Intuitive navigation
- Real-time visual feedback
- Responsive design for multiple devices

## User Interface Screens

| Screen          | Description                                     |
|-----------------|-------------------------------------------------|
| Home Page       | Displays product categories and recommendations |
| Try-On Screen   | Shows live AR visualization                     |
| Login/Register  | Handles user authentication                     |
| Profile Page    | Displays user history and preferences           |
| Screen          | Description                                     |
| Admin Dashboard | Provides management tools and analytics         |

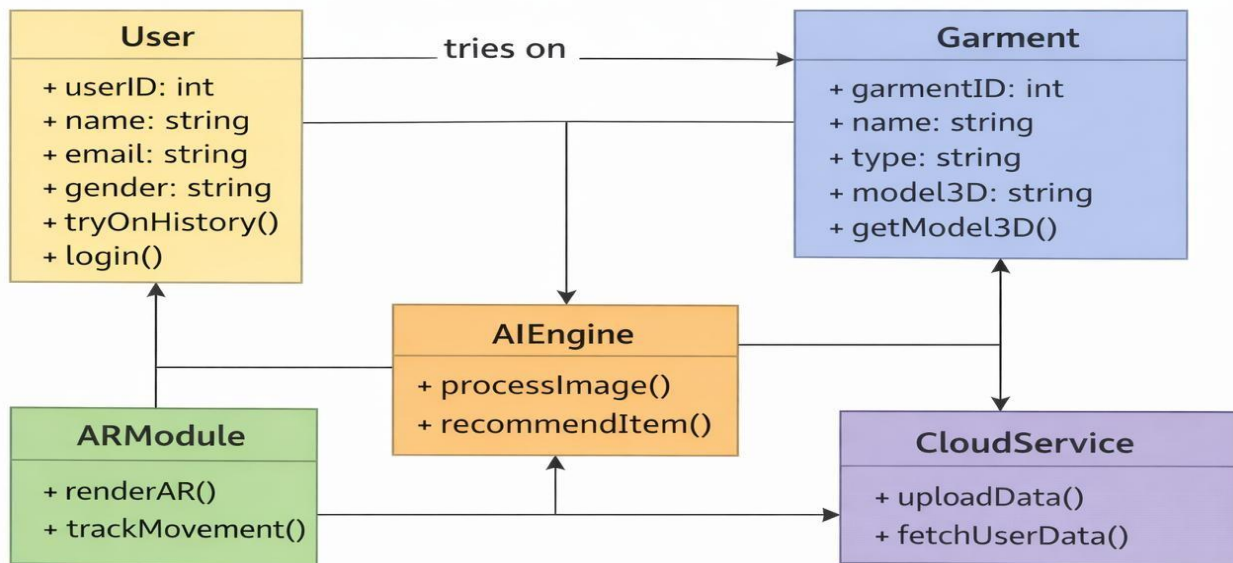
## Input Validation and Error Handling

- Mandatory field checks
- Format validation for email and passwords
- User-friendly error messages

## UML Diagrams Description

### Class Diagram

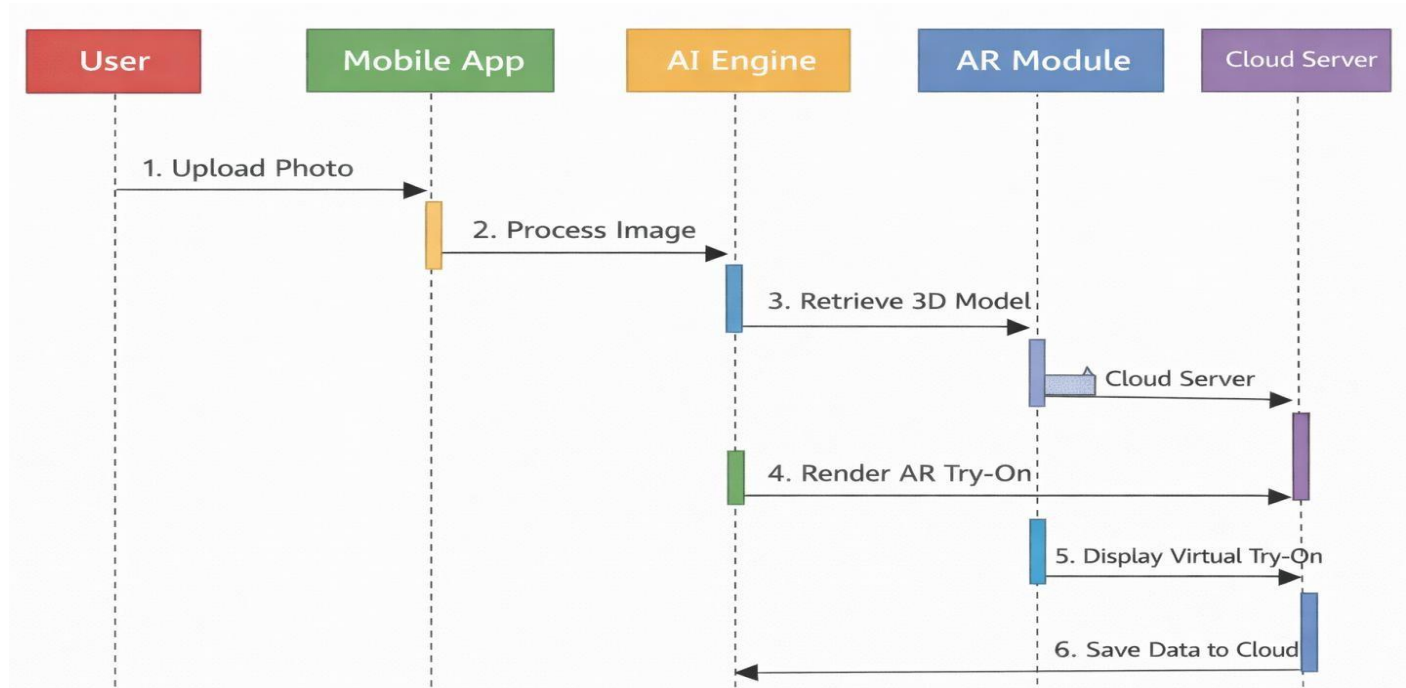
The class diagram illustrates the system structure, including classes such as User, Product, Tryon Session, AI Engine, and AR Module, along with their attributes, methods, and relationships.



**Fig.1 : Class Diagram**

## Sequence Diagram

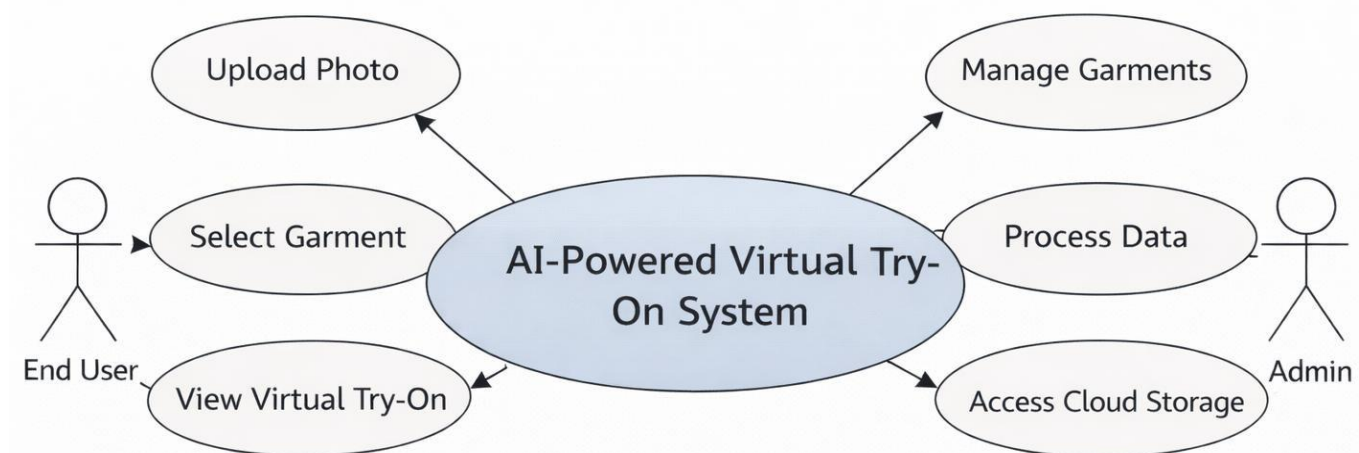
The sequence diagram represents the flow of interactions during a virtual try-on session, starting from user input to AI processing, AR rendering, and result display.



**Fig.2 : Sequence Diagram**

## Use Case Diagram

The use case diagram identifies interactions between actors (End User and Admin) and the system, highlighting major functionalities such as uploading images, selecting products, managing data, and accessing cloud storage.



**Fig.3 : User Case Diagram**

## **Security Design**

### **Authentication and Authorization**

- Secure login mechanisms
- Role-based access control
- Token-based authentication for APIs

### **Data Security and Privacy**

- Encrypted data transmission
- Secure cloud storage
- User consent for image processing

## **Assumptions and Constraints**

### **Assumptions**

- AI models are pre-trained and optimized
- Cloud infrastructure is reliable and scalable

### **Constraints**

- Network latency in low-bandwidth environments
- Device compatibility limitations

## **Future Enhancements**

- Integration with e-commerce payment systems
- Support for additional product categories
- Improved AI accuracy through model retraining
- Multi-language user interface support

## **References:**

- [1] MediaPipe Face Detection Documentation, Google, <https://mediapipe.dev/>
- [2] Three.js 3D Library Documentation, <https://threejs.org/docs/>
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- [8] IEEE 830-1998 Standard for Software Requirements Specification

## **Appendix A: Glossary**

**AR (Augmented Reality):** Technology that overlays digital content onto the real world view.

**Computer Vision:** Field of AI that enables computers to interpret visual information from images or video.

**Face Landmark:** Specific points on a face (eyes, nose, mouth corners) used for tracking and positioning.

**GLB/GLTF:** Standard file formats for 3D models used in web applications.

**JWT (JSON Web Token):** Secure method for transmitting authentication information.

**MediaPipe:** Google's framework for building multimodal machine learning pipelines.

**Three.js:** JavaScript library for creating 3D graphics in web browsers.

**WebRTC:** Web Real-Time Communication, enabling real-time video/audio in browsers.

**WebSocket:** Protocol for real-time bidirectional communication between client and server.